IAS Physics

Dictionary

Topic 1: Mechanics

Centre of Gravity

The point at which the weight of an object appears to act.

Equilibrium

Both resultant force and resultant moment around any point must equal zero.

Newton's First Law (N1)

An object will remain stationary or continue to move at a constant velocity unless acted on by a resultant force.

Newton's Second Law (N2)

The resultant force applied to an object is porportional to the object's rate of change in momentum.

OR

Resultant force is equal to mass multiplied by acceleration.

Newton's Third Law (N3)

When an object A exerts a force onto object B, object B will exert an equal and opposite force onto object A.

OR

There is an equal and opposite reaction to every action.

Scalar Quantity

A quantity with only magnitude, no direction.

The Principle of Conservation of Linear Momentum

Total momentum remains constant in a closed system.

OR

The sum of momenta before a collision is equal to the sum of momenta after a collision, provided no external forces are acting.

Terminal Velocity

The maximum velocity of an object when forward forces and frictional forces are equal.

Vector Quantity

A quantity with both magnitude and direction.

Topic 2: Materials

Breaking Stress (Fracture Stress)
The value of stress within a material when it breaks.

Elastic Deformation
A deformed object will return to its original shape and size once the force is removed.

Elastic Limit
The point of maximum stress applied to an object before it is permanently deformed.

Hooke's Law
An elastic object's extension will be proportional to the force applied to it, until the limit of proportionality is reached.

Laminar Flow
Layers of fluid move parallel to one another without mixing between layers.

Limit of Proportionality

The point after which Hooke's law no longer applies to an object; strain/extension does not increase proportionately to force/stress.

Plastic Deformation

A deformed object will **not** return to its original shape and size once the force is removed.

Turbulent Flow

Layers of fluid mix together in an unpredictable manner.

Upthrust

Upthrust is equal to the weight of fluid displaced by a submerged object.

Ultimate Tensile Stress

The highest possible stress within a material.

Yield Point

The point at which a material has a large increasse in extension for a small force.

Young Modulus

The ratio of stress to strain for a given material.

Units

Base Units

Basic Quantity	Unit Name	Unit Symbol
Time	Second	**s**
Length	Metre	**m**
Mass	Kiogram	**kg**
Temperature	Kelvin	**K**
Amount of Substance	Mole	**mol**
Light Intensity	Candela	**cd**
Electric Current	Ampere	**A**

Derived Units

Derived Quantity	Unit Name	Unit Symbol	Base Units Equivalent
Force	Newton	***N***	**kg m/s²**
Energy (Work)	Joule	***J***	**kg m²/s²**
Power	Watt	***W***	**kg m²/s³**
Frequency	Hertz	***Hz***	**/s**
Charge	Coulomb	***C***	**A s**

Voltage	Volt	***V***	**kg m² / s³ / A**
Resistance	Ohm	**Ω**	**kg m² / s³ / A²**

Power Prefixes

Name	Factor	Symbol
Deca-	x10^**1**	***da***
Hecto-	x10^**2**	***h***
Kilo-	x10^**3**	***k***
Mega-	x10^**6**	***M***
Tera-	x10^**12**	***T***
	Negative Powers	
Deci-	x10^**-1**	***d***
Centi-	x10^**-2**	***c***
Milli-	x10^**-3**	***m***
Micro-	x10^**-6**	***μ***
Nano-	x10**-9**	***n***
Pico-	x10^**-12**	***p***